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Variation in degrees of Prosocial and Risk-Taking Behavior of MOBA Gamers

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Abstract

With Multiplayer Online Battle Arena (MOBA) becoming an increasingly popular genre of online games, affecting thousands and millions of players worldwide, we investigate into how these games actually affect our day-to-day experiences. More exactly, we look at its definite impacts on player (prosocial and risk-taking) behavior with respect to their preference for character gaming roles (tank, carry, and support). Participants are composed of college students (MOBA and non-MOBA players) from different universities within Metro Manila. Overall findings showed that no significant difference exists between the levels of prosocial, as well as risk-taking, behavior of MOBA players with respect to the three character gaming roles. The study, however, also indicated that a significant difference exists between Non-MOBA and MOBA participants with respect to their prosocial and risk-taking behavior— where tanks (MOBA) were found to exhibit significantly higher levels of prosocial behavior and all three character gaming roles (MOBA) identified as having significantly higher levels of risk-taking behavior as compared to Non-MOBA participants. This further supports a related claim of an underlying phenomenon called Proteus effect where players infer their expected behavior and attitudes from observing their avatar's appearance.

Keywords: *Multiplayer Online Battle Arena (MOBA), Avatar, Tank, Carry, Support, Prosocial behavior, Risk-taking behavior, Proteus effect.*

The Relationship between Prosocial and Risk-taking Behavior and Character Role among the Players of MOBA

In personal computer consoles, games have taken many forms and genres. One of which are *MMOG*s or Massively Multiplayer Online Games. Accordingly, *MMOGs* can be classified in several ways depending on the system environment functions and player-plot factors. To specify, some sub-genres of MMOG include: *MMORPG* (Role-Playing), *MMOFPS* (First-Person Shooter), *MMOSG* (Strategic), and many more. However, aside from these games, a new sub-genre of *real-time strategy* game (RTS) called Multiplayer Online Battle Arena, more fondly referred to as MOBA, has increasingly been popularized by a game modification called *DotA* (Defense of the Ancients). Tracing from its successful release in 2003, a wide variety of game modifications under the MOBA category were subsequently developed. For instance, games such as League of Legends (LoL) in 2009, Heroes of Newerth (HoN) in 2010, Guardians of Middle Earth (GoME) in 2013, and the stand-alone sequel Dota 2, of the same year, later surfaced (Game Debate, 2013; Yang, Harrison, & Roberts, 2014).

What is MOBA?

There are several gameplay elements to describe MOBA. The key feature presents that there are two opposing teams whose collective goal is to destroy the enemy's base to win. However, defensive towers as well as weak, periodically-spawned units are placed along lanes to delay opponents from permeating allied defenses. In MOBA, players control a single powerful in-game avatar typically referred to as *hero*. Whenever a hero kills an enemy unit, it gains experience points allowing it to level up, gain new abilities, and in the most basic essence- become more potent (Kolwankar, 2012). MOBA games are highly dynamic in team-based combat. Group strategy likewise requires a collaborative involvement among the members of the team. However, in this study, the focus will not be on group strategy. Rather, the

emphasis will be on the individuality of players in relation to the characteristics of their controlled heroes. Hence, different character role elements will further be discussed in the subsequent sections of this study.

GAMING ROLES

As discussed by Yang, Harrison, and Roberts (2014) and in the official *Dota 2* site (Dota 2, n.d.), there are three basic types of gaming roles: *tank*, *carry*, and *support*.

Having strength as their main attribute, *tanks* are typified as heroes having the most health points and characteristic of relatively high damage points as well. They are mostly durable and can sustain huge amounts of damage. Commonly referred to as *setters*, the tanks' main job is to frontier team clashes to relieve their team mates from receiving a great deal of damage. In most situations, tanks also initiate *ganking*. Ganking (gang-killing) is usually used in the context of an ambush or the act of a player or group of players roaming and killing enemy heroes. By doing so, they are able to defeat opponent heroes in the fastest and most efficient way as possible. In some cases, they also disrupt the opponent's strategic placement in clashes by doing area skills. The second type of gaming role is *support*, whose function is to supply team necessities. Generally, supports' main attribute is intelligence. As the name implies, *supports* enhance the team's efficiency by providing health, attribute, or ability boosts by using support skills and potions. One of their main roles is to make sure that their team mates survive clashes. Supports can also inflict occasional damages or decreases in enemy strength. Heroes of this type also frequent the use of wards to widen the team's visual scope in the map or even reveal invisible enemies. The idea of support as an all-item, protection provider and no damage, however has changed. Modern variations in item-building and skill modifications allowed for more flexible role classification hence more complex terms such as "tank-support" or "jungle-carry." But just like in the

case of tanks, supports have strong initiatives to commence clashes (R. J. Tan, personal communication, June 19, 2014). Lastly, *carry* types, oftentimes characterized as *hitters*, are players who function as heavers. They inflict heavy damage on the opposing team by maximizing their damage points in order to deal constant detriment in team combat. Carries have agility as their central attribute. In early game periods, the *carry* types are expected to excel in item and level or ability acquisition. However, heroes of this nature are likely to have lower health points as compared to the two other roles. And unlike tanks and supports that can fuse together, R. J. Tan clarified that "a carry is just a carry. There's no thin line" (personal communication, June 19, 2014).

Albeit individualized in the context of heroes, MOBA games are interactive and social in nature. Cooperation and coordination have, likewise, become necessities in the dynamics of MOBA gameplay. In instances such as an ambush, members constantly collaborate for a strategically planned goal of killing an opponent hero. The plan could include risking the life of a tank hero while the carry executes a skill set and a support timelessly boosting his teammate. This emphasizes a study that suggests that the social context of experiential gaming and the concept of how a game is played may be of bigger importance with respect to its contents (Jerabeck and Ferguson, 2013). As these users engage in playing the game, they tend to manifest their own personality through their in-game behavior. But before we delve further into personality and gaming behavior, let us first examine the user-avatar relationship in these games.

USER-AVATAR RELATIONSHIP

The Avatar

WHAT IS AN AVATAR? Players can live a virtual life by means of their avatar and, dependent on the nature of the game, use their avatar to explore, compete, create content, collaborate, or socialize. An *avatar*, as

defined by Hollingshead and Poole (2012), is a graphical embodiment of a participant that conveys the participant's identity, presence, location, and activities to others; and as defined by Ruyter and Conroy (2002), as cited in Suh (2013), a virtual self is the configuration of the defining characteristics of a person in a virtual space. Where Gee (2003), as cited in McCreery, Krach, Schrader and Bone (2012), also suggested that the virtual self has become a new component of one's overall identity, existing in conjunction with the public and private persona. The avatar has been described as an extension of oneself, replaced with all necessary psychological components of the physical equivalent (McCreery et al., 2012) and therefore, connects the avatar to the physical body and through this interaction of one's online and offline identities, the avatar may become important to oneself. Hayles (1999), as cited in Behm-Morawitz (2013), argues that the virtual is never independent of the body. And to a certain extent, some individuals even feel and think that their own selves are the ones interacting with the virtual world (Yee, Bailenson, & Ducheneaut, 2009).

The Proteus Effect and Self-Representation Embodiment

Proteus Effect

Further findings have suggested evidences for user-avatar relationship with respect to MOBA: one of which is *Proteus effect*. Studies by Yee, Bailenson, and Ducheneaut (2009) have shown that people infer their expected behaviors and attitudes from observing their avatar's appearance; a phenomenon known as Proteus effect. It was found that different characteristics of avatars in an online game have significant effects on player's performance. According to Yee and Bailenson (2009), the Proteus Effect has shown that the physicality of an avatar appearance may already lead to behavioral changes in users. These studies suggest that embodiment plays an important role in the Proteus effect. Consequent findings by Fox, Bailenson, and Tricase (2013) supported the notion of Proteus effect wherein

participants who wore sexualized avatars internalized the avatar's appearance and self-objectified, reporting more body-related thoughts than those wearing non-sexualized avatars.

Embodiment of Self-Representations (Avatars)

When a user embodies his self-representation, he observes himself behaving in the virtual form, and formulates inferences regarding his internal beliefs or attitudes based on these observations. After the embodiment, the user's behavior then conforms to the modified self-representation regardless of the true physical self (Yee & Bailenson, 2007; Yee, Bailenson, & Ducheneaut, 2009). A related study by Mosley (2010) examined the relationship between player's personality characteristics and their online behavior; including character faction, class selection, and game play in the massively multiplayer online role playing game. Statistical analysis subsequently showed that there were significant relationships between personality traits and engagement in player-versus-player gameplay. These parallels the attachment players develop to the characters they portray, and the transference players experience from their game world interaction into their out-of-game lives.

PERSONALITY PATTERNS

The emphasis of this research is to identify personality patterns of players of MOBA in connection with their preference for the three types of gaming roles. Furthermore, the study has focused on two particular personality traits which can be portrayed through the MOBA gameplay: *prosocial* and *risk-taking behavior.* Following this discussion, the study hypothesizes two assumptions. The first hypothesis states that there would be a significant difference between MOBA and Non-MOBA players in relation to their prosocial and risk taking behavior. The second assumption is that there would be a significant difference among the levels of prosocial and risk-taking behavior of the three character gaming roles. The notion is that

players who often take on the role of a tank are characterized as having average prosocial behavior and high risk-taking behavior, while players who take on the support role are high on prosocial behavior and average on risk-taking behavior, and finally, the carry role as being low on both.

Prosocial Behavior

Prosocial behavior, as defined by Eisenberg and Fabbes (1990), is a voluntary behavior intended to benefit another. General acts of generosity and helpfulness such as assisting a friend in need, donating toys to charities, and cooperating in group projects may exemplify qualities of prosocial behavior where individuals and groups, benefit. In connection with MOBA games, prosocial behaviors are exhibited as players cooperate with one another to achieve a unified goal of winning the game. However, winning is but only one of the many shared aims in playing team-oriented games. Not only do these games provide competitive fulfillment, but also, companionship and satisfaction in social interactions (Fang & Zhao, 2010). Like in the case of character roles, the propensity of players to choose tank may actually be indicative of their prosocial attribute.

Risk-Taking

Yoe (2012) defined *risk* as a situation or event in which something of human value, including human themselves, is put at stake and where the outcome is uncertain. It is the chance of an undesirable outcome and the outcome can be a potential loss or a potential gain. Additionally, risk-taking behavior may be defined as the engagement of an individual in behavior that may have potentially harmful outcomes (Ernst et al., 2013); or as Magar, Phillips, and Hosie (2008) defined risk-taking— it is the voluntary participation in any behavior which carries some probability for negative consequences. Risk perception, in general, explains the processing of physical signals or information about potentially harmful events or activities, and the formation of a judgment about seriousness, likelihood, and

acceptability of the respective event or activity. When a person has an increased risk perception, he/she is more likely to engage in risky behavior (Kern et al., 2013).

Methodology

Participants

A total of 695 college students, aged 16 to 21(*M*=18.13, *SD*=1.22), from five universities within Metro Manila participated in the study. Four hundred sixteen (60%) were identified as non-MOBA players while 279 (40%) were MOBA players.

Design

The proponents made use of descriptive design as an overall research framework. The variables considered in the study were the character gaming roles consisting of three factors: tank, carry, and support; while the second variable considered were the prosocial and risk-taking behavior of the participants. The design of the study helped identify if a significant difference exists among the prosocial behavior scores of tank, carry, and support, and their risk-taking behavior scores as well.

Materials

The researchers utilized three instruments for gathering the data needed in the study. The following are the instruments and their descriptions:

1. **MOBA Character Role Questionnaire -** MOBA questionnaire is a 16-item, multiple-choice test designed by the researchers that showcases traits specific to the character roles (tank, carry, and support) in MOBA games to identify the preference of the respondent.
2. **DOSPERT Scale -** Domain-Specific Risk-Taking (DOSPERT) Scale is a 40-item psychometric scale developed by Weber, Blais, and Bentz (2002) that assesses risk taking in five content domains*: financial decisions*, *health*

and safety, *recreational*, *ethical*, and *social decisions.* The Risk-Likelihood measure was used in the study.

3. **Prosocial Personality Battery (PSB) -** PSB is a 56-item personality test developed by Penner, Fritzsche, Craiger, and Freifeld (1995) that measures prosocial personality orientation. The test consists of two domains: Other-Oriented Empathy and Helpfulness.

Procedures

Data were collected by conducting a battery of tests to college students selected through purposive and convenience sampling methods. Specifically, the targeted population of the study was MOBA players. The participants were given a test booklet containing the PSB, DOSPERT, and MOBA questionnaire, respectively, as well as a separate answer sheet. The test administration roughly took 30 to 40 minutes of the participants' time while the MOBA questionnaire consumed an approximated time of five to 10 minutes. All of the participants were required to answer the PSB and the DOSPERT while the MOBA Questionnaire was answered by the MOBA players only. MOBA players were identified through the question: whether they play MOBA or not, which was placed after the DOSPERT and before the MOBA questionnaire starts in the booklet. Participants were also asked to name other MOBA players in their class so as to validate claims of being a MOBA player.

At the end of every test administration, answer sheets of MOBA and Non-MOBA participants were segregated. Furthermore, data gathered from the MOBA players were divided with respect to their preferred character roles. Roles were subsequently identified based on their answers in the MOBA questionnaire, in which of the three roles they preferred the most.

Data Analysis

After data gathering, incomplete and invalidated answer sheets were discarded from the protocols. Data were scored and recorded, separating the

participants who play MOBA from students who do not. Both personality tests and the MOBA answer sheets were checked. In obtaining statistical data, raw scores were computed using Microsoft Office Excel 2013 Data Analysis Toolpack while IBM SPSS Statistics 20 was used to support data resulting from the use of Microsoft Office Excel.

A frequency distribution table was used to determine the spread of scores among Prosocial and Risk-taking behavior of all the participants (MOBA and Non-MOBA). Data obtained from Non-MOBA participants was used as a baseline between the two variables. Consequently, Descriptive Statistics was used to determine the means, medians, and standard deviations of the participants. A Two-Sample t-test was then utilized to assess whether a significant difference exists between Non-MOBA and MOBA participants in terms of Prosocial and Risk-taking behavior. On the other hand, One-Way Between Subjects ANOVA was used to examine whether the three character roles differ in terms of Prosocial and Risk-taking behavior.

Results and Discussion

Analyses of data were conducted using the Data Analysis Toolpack from the software Microsoft Excel 2013 and IBM SPSS Statistics 20. All computations are at the 0.05 level of significance.

Difference between Non-MOBA and MOBA participants in terms of Prosocial and Risk-taking Behavior.

A Two sample t-test was used to assess whether a significant difference exists between Non-MOBA and MOBA participants in terms of Prosocial and Risk-taking behavior. The results showed no significant difference between Non-MOBA participants (M = 189.23, SD = 15.67) and MOBA participants (M = 190.72, SD = 15.75), $t(594) = -1.22$, $p > .05$ in terms of Prosocial behavior. On the other hand, the results showed significant difference between Non-

moba participants (*M* = 100.40, *SD* = 17.96) and MOBA participants (*M* = 108.20, *SD* = 20.83), *t*(535) = -5.11, *p* < .05 in terms of Risk-taking behavior. Table 1 shows the means and standard deviations between Non-MOBA and MOBA participants in terms of Prosocial and Risk-taking behavior.

Table 1

Means and Standard Deviations of Non-MOBA and MOBA in terms of Prosocial and Risk-taking Behavior

	PROSOCIAL BEHAVIOR *M(SD)*	RISK-TAKING BEHAVIOR *M(SD)*
NON-MOBA	189.23 (15.67)	100.40 (17.96)
MOBA	190.72 (15.75)	108.20 (20.83)*

Note. * p < .05.

Further analysis revealed that Tank (*M* = 192.84, *SD* = 14.83) showed significant difference from Non-MOBA participants (*M* = 189.23, *SD* = 15.67), *t*(125) = -2.02, *p* < .05 in terms of Prosocial behavior. Similar analysis revealed that the three character roles also showed significant difference from Non-MOBA participants in terms of Risk-taking behavior. Table 2 shows the means and standard deviations between Non-MOBA and the three Character Gaming Roles in terms of Prosocial and Risk-taking behavior.

Difference among Character Gaming Roles in terms of Prosocial and Risk-taking Behavior.

A One-Way between subjects ANOVA was used to assess whether the three character roles differ in terms of Prosocial and Risk-taking behavior. The results showed no significant difference between the three character gaming roles: Tank (*M* = 192.84, *SD* = 14.83), Carry (*M* = 188.14, *SD* = 15.32), and Support (*M* = 192.01, *SD* = 16.89), *F*(2,278) = 2.57, *p* = 0.08 in terms of Prosocial behavior. Similarly, there was no significant difference

between the three character gaming roles: Tank (*M* = 108.99, *SD* = 18.15), Carry (*M* = 109.28, *SD* = 21.82), and Support (*M* = 105.95, *SD* = 22.08), *F*(2,278) = 0.69, *p* = 0.50 in terms of Risk-taking behavior. Table 2 shows the means and standard deviations between Non-MOBA and the three Character Gaming Roles in terms of Prosocial and Risk-taking behavior.

Table 2

Means and Standard Deviations of Non-MOBA and Character Gaming Roles in terms of Prosocial and Risk-taking Behavior

	PROSOCIAL BEHAVIOR *M(SD)*	RISK-TAKING BEHAVIOR *M(SD)*
NON-MOBA	189.23 (15.67)	100.40 (17.96)
TANK	192.84 (14.83)*	108.99 (18.15)*
CARRY	188.14 (15.32)	109.28 (21.82)*
SUPPORT	192.01 (16.89)	105.95 (22.08)*

Note. * p < .05.

General Discussion

One of the goals of this research is to find out if a significant difference exists in prosocial as well as risk-taking behavior of MOBA players with respect to the three character gaming roles. The assumptions, therefore, of the study are that: people who frequent themselves with the use of the tank role will have high and average scores on risk-taking and prosocial behavior, respectively; support-users will score high on prosocial behavior and average on risk-taking behavior; and finally, the carry-users will score low on both personality variables. The study, however, revealed that no significant differences exist among the three character roles as they were compared, separately, to the two personality variables.

According to Eisenberg and Fabbes (1990), prosocial behavior is a voluntary behavior intended to benefit another. Such behavior is manifested

in various ways relative to the person's altruistic disposition. In virtual games, prosocial behavior may be displayed in numerous way such as helping a teammate during clashes or ganks, providing of group necessities, and other more generally cooperative and helpful acts. Even simple expressions of verbal tutoring of skillful gamers in response to their frustration for less-skilled teammates may be classified as a display of prosocial behavior (Jerabeck & Ferguson, 2013).

In relation to this research, it was found out that the three character roles did not differ significantly with respect to the prosocial measure of behavior. A probable explanation to this may be that, the concept of helping behavior in the context of games such as MOBA is too vaguely measured in the sense that, cooperation or prosocial behavior has become an innate characteristic of players. Teamwork, cooperation, and strategic planning altogether have become fundamental points of the game and therefore, all may have the same extent of capacity to exhibit this helping behavior.

As the prosocial behavior was related to these roles, the second assumption was that each role may also manifest risk-taking behavior since making an effective hero is not an easy task and playing a combat-based game involves an assessment of and engagement in a risky situation.

Yoe (2012) defined risk as a situation or an event in which something of human value, including humans themselves, is put at stake and where the outcome is uncertain. The outcome is never known until it is already experienced by the person. Additionally, Ernst et al. (2013) highlighted that the engagement of an individual in a risk-taking behavior may have potentially harmful outcomes. This means that engaging in a risky activity yields to either positive or negative results. In the case of MOBA games, positive outcome refers to winning, while the negative outcome is depicted through losing. The three character roles, however, did not again differ in terms of their risk-taking behavior; which means that all of the MOBA players

who participated in this study, regardless of their inclination for a specific gaming role, manifested the same level of risk-taking behavior. A likely explanation for this may be that, again, the game necessitates that all engage in a risky behavior to the extent that risk-taking has become a constant state for the players. For instance, a team mate may be in a near-death situation in a team assault while another member could save this hero by risking his own. Implications like these provide basis as how risk-taking behavior could then be displayed by any or every user in the group.

This leads to another speculation that the reason why there are no significant differences on their risk-taking or prosocial behaviors is because of the uncertainty of the strategy their team employs through each game. Hence, difficulty measuring specific variables arises as relative changes in the measurement of these variables occur dynamically.

On another perspective, the results of this research, by some means, support the study of Jerabeck and Ferguson (2013) that discovered the implication of social background of game experience and the suggestion that how the game must be played may possess a greater role than what the game content actually is. That is to say, players are required to be flexible and adaptable when paired with various teammates and in anticipation of the different sets of heroes in the opposing team as well. Hero choice is a vital part of each game as every hero has its own strengths and weaknesses. For a team to work effectively, the three roles must complement each other; maximizing their capacities and fulfilling their tasks both separately and interdependently. For the opposing team's perspective, they view their avatars and the other team's group as contradictory. For example, a hero that has an invisibility skill may be selected by one team. Given this situation, the other team will have the option to choose a hero who has the ability to reveal invisible units. Taking this into account, regardless of the gaming role,

the player must adapt to the chosen hero for them to refute the opposing team that, in turn, gives them advantage.

Another explanation may be that: If there are too many heroes of one role in a group's roster, and even though another player prefers that same role, this player may be forced to select another hero aside from what he prefers so as to prevent repetition of tasks.

Skilled players were indirectly taught to be flexible in role selection in order for them to have a higher probability of winning the game. On the other hand, the less skilled players eventually learn to be just as flexible in playing alternative roles as they play the game more constantly.

Another objective of the research is to find out if there are significant differences between the risk-taking and prosocial behavior of MOBA and Non-MOBA participants. All character roles (MOBA) were found to be significantly different from Non-MOBA participants in terms of risk-taking behavior while the tank character role (MOBA) is the only role that was found to be significantly different compared to Non-MOBA participants in terms of prosocial behavior.

A plausible explanation behind these results may be explained by an underlying phenomenon called Proteus effect which states that when a player is exposed to a virtual game that involves different characters, this allows the player acquire his avatar's characteristics as well. According to Yee, Bailenson, and Ducheneaut (2009), Proteus effect is a phenomenon wherein people infer their expected behaviors and attitudes from observing their avatar's appearance. Furthermore, Proteus effect conveys that the physical aspects of an avatar may lead to behavioral changes among users (Yee & Bailenson, 2009). But beyond the relative physicality of avatars in-game, a more intriguing aspect of the study supposes that a possible reason why MOBA participants scored relatively higher on specific personality scales is because they have a tendency to internalize the character roles

they select. Ergo, innate characteristics of a gaming role such as prosocial and risk-taking behavior may actually affect their “offline,” real-life behavior.

The study, therefore, does not support the concept of the virtual self where the person sees his avatar as an extension of his true self. As stated by McCreery et al. (2012), the virtual self has been described as a copy of, or an extension of oneself, replaced with all necessary psychological components of the physical equivalent. From the information we gathered, the research revealed that participants did not see their real selves on their heroes; rather, they saw their avatars as a reflection of what the situation requires them to be.

As a final point, the study stresses out two simple things. First, the situation of their game play itself affects how a person behaves in a virtual world regardless of their character role preference, particularly in competitive and cooperative games. And second, the avatar itself also affects the user behaviorally. Even though the physicality of the avatar does not translate into psychological changes, the characteristics of its role affect the player. These findings subsequently conclude that online behavior can actually affect real-time behavior of players, in general.

The scope and limitations were also included to purposefully define the research further. Firstly, the study involved adolescents of both genders. The findings, therefore, should not be generalized with other age groups. The researchers also did not consider the participants’ longevity in playing the game as it was viewed as an insignificant factor in identifying their personality patterns in the virtual world. In qualifying participants as MOBA players, they simply were described as those who play at least any single one of the games aforementioned: DotA, Heroes of Newerth, League of Legends, or Dota2. The participants also restricted to university students within Metro Manila.

Alternatively, the study offers a few key implication. Since the results have revealed that there is no significant difference in the prosocial and risk-taking behavior among the three character roles, subsequent researches can then further investigate other player behavior in the virtual world. Personality and group dynamics such as flexibility, adaptability, teamwork, and cooperation are just some of the many variables that can be researched on regarding the dynamics of playing. Moreover, future works can add more in-depth qualitative information about the players of MOBA, MMORPG (Massive Multiplayer Online Role-Playing Game), or virtual gaming in general. Since this study only involves MOBA players, a research on Persistent Virtual Environments (PVEs) or MMORPG could then be considered since avatars of this genre can be modified or altered by players themselves. The modifiability factor of avatars present in MMORPG gameplay which is not applicable in MOBA can subsequently provide future researchers more information regarding personality formation, behavioral translation, and the like.

In addition, by utilizing the mechanics of the different fields of gaming, inquiries can be made with regards to the distinctions among personalities of the gamers who play different games. These studies could further probe to desired or idealized self-representations of players as expressed in MMORPG, aggression or violence for people who constantly play First-Person Shooter games, or strategic dynamism in Real-Time Strategy. Researches on these areas additionally promotes the encompassing implications of psychology in the real world; an application of Psychology in the social schemes of both real-time and virtual (online) worlds.

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